

RAG Pipeline — Model Comparison Report

Models evaluated: gemma3:4b, gemma4:e2b, minstral-3:3b, llava-phi3:3.8b, qwen3.5:2b | Generated: 20 April 2026, 09:19 PM

Retrieval Metrics (shared — model-independent)

Metric	gemma3:4b	gemma4:e2b	minstral-3:3b	llava-phi3:3.8b	qwen3.5:2b
RETRIEVAL					
M1 Embedding Time ↓	14.9993 s				
M2 Index Size ↑	176 vectors				
M3 Retrieval Latency ↓	0.1676 s				
M4 Cosine Sim (Text) ↑	0.7436				
M4 Cosine Sim (Image) ↑	0.3797				
M5 Page Coverage@k ↑	89.29 %				
M9 Context Length ↑	3255 chars				

Generation Quality Metrics (per model, vs ground truth)

GT available: 14/14

Metric	gemma3:4b	gemma4:e2b	minstral-3:3b	llava-phi3:3.8b	qwen3.5:2b
GENERATION QUALITY					
M6 ROUGE-1 ↑	0.4814	0.4970	0.4674	0.4564	0.4616
M7 ROUGE-2 ↑	0.1581	0.1846	0.1385	0.1328	0.1585
M8 ROUGE-L ↑	0.2508	0.2809	0.2225	0.2419	0.2543
M10 BLEU ↑	0.0811	0.1207	0.0529	0.0725	0.0938
M11 METEOR ↑	0.3358	0.3411	0.3445	0.3184	0.3443
M12 BERTScore (F1) ↑	0.8688	0.8735	0.8648	0.8606	0.8695
M13 FCD ↓	15.29	16.22	16.65	17.30	16.73
M14 Faithfulness ↑	83.90 %	87.97 %	63.99 %	77.65 %	73.23 %
M15 GT Coverage ↑	40.41 %	37.62 %	42.18 %	36.24 %	39.08 %

Performance Metrics (per model)

Metric	gemma3:4b	gemma4:e2b	ministral-3:3b	llava-phi3:3.8b	qwen3.5:2b
PERFORMANCE					
M16 E2E Latency ↓	35.0156 s	42.9806 s	128.1259 s	47.8833 s	67.4074 s
M17 Throughput ↑	0.029 q/s	0.023 q/s	0.008 q/s	0.021 q/s	0.015 q/s
M18 CPU Usage ↓	1.68 %	1.42 %	0.86 %	2.68 %	2.21 %
M19 RAM Usage ↓	0.785 GB	0.125 GB	0.309 GB	0.746 GB	0.584 GB
GPU Usage	30.05 %	1.01 %	0.49 %	6.76 %	1.58 %